MSE 230 *Structure and Properties of Materials* (Spring 2026)
School of Materials Engineering

Course Description: Credit Hours: 3.00. The relationship between the structure of materials and the resulting mechanical, thermal, electrical, and optical properties. Atomic structure, bonding, atomic arrangement; crystal symmetry, crystal structure, habit, lattices, defects, and the use of X-ray diffraction. Phase equilibria and microstructural development. Applications to design**.**

Course Objectives for MSE 230 can be found at:
https://engineering.purdue.edu/MSE/Academics/Courses/MSE230/index.html

1. To provide an introduction to Materials Science and Engineering (MSE) as the study of structure-properties-processing relationships in engineering materials (metals, polymers, ceramics, and composites).
2. To develop the general and specific relationships between the different levels of internal structure (atomic, molecular, crystal, grain, etc.) and the basic properties (physical, mechanical, thermal, electrical, etc.) of engineering materials.
3. To show how structure, and thus properties, are controlled and manipulated in the basic processing operations.
4. To apply the structure-properties-processing approach to the specification of materials for engineering applications and develop appreciation for MSE as a tool in engineering design.

Specific Goals for the Course

1. All Students

A. Recognize basic MSE nomenclature, basic microstructure, associate terms with the appropriate structure/phenomena, and be able to differentiate between related structures/phenomena. Examples:

- FCC and BCC crystal structures.
- Ionic and Covalent crystal structures.
- Elastic and Plastic deformation.
- Isomorphous and eutectic phase diagrams.
- Interstitial and vacancy diffusion mechanisms.
- Polymers exhibit a distribution of molecular weights.
- Identifying and drawing crystallographic planes and directions.

B. Perform simple calculations to quantify material properties and microstructural characteristics. Examples:

- Interplanar spacing of a family of atomic planes given a x-ray diffraction pattern.
- Equilibrium vacancy concentration at a given temperature.
- Dimensional changes associated with elastic and plastic deformation.
- Fracture strength for a given flaw size.
- Phase composition and fraction using phase diagrams.
- Apply Reuss and Voight models to determine modulus of composites.

C. Recognize the effect of composition and microstructure on material properties. Examples:

- Hall-Petch effect
- Dislocation density effect on yield strength and electrical conductivity.

- Alloying effect on yield strength and electrical conductivity.
- Inverse relationship between yield strength and fracture toughness.
- Correlation between type of atomic bonding and the mechanical and electrical properties of different classes of materials.

2. Most Students

A. Take information from a known situation and apply it to a new situation. Examples:
- Effect of temperature and applied stress on the peak positions in a XRD pattern.
- Simple calculations with multiple steps.

B. Predict property response or microstructural changes based on imposed conditions. Examples:
- Predicting microstructure using a phase diagram.
- Predicting microstructure using a TTT diagram.
- Effects of temperature and alloying on the resistivity of metals.

C. Assess the interplay of two material properties. Examples:
- Effect of atomic bond strength on Young's modulus, coefficient of thermal expansion and melting temperature.
- Determine if a material will yield or rupture at a given applied stress.

Relation of Course to Student Outcomes:
(MSE-1, ABET-1) an ability to identify, formulate, and solve complex materials engineering problems by applying principles of engineering, science, and mathematics.

TEXTBOOK

We are using a new form of the classic MSE 230 textbook (Introduction to Materials Science by Callister) that provides real-time feedback on your learning. *This virtual textbook is required as you will have weekly assignments (**due Sunday night by 11:59 PM**) to complete.* These assignments are part of the interactive book. You can always get a perfect score on these daily assignments. The book cost is $69. You can access the book on December 22, 2025; your subscription lasts until May 29, 2026. The cutoff date is April 27 (last day students can subscribe). If you haven't already done so, follow the instructions below to access the book.

1. Sign in or create an account at learn.zybooks.com
2. Enter zyBook code: PURDUEMSE23000Spring2026
3. Subscribe

* zyBook name: MSE 23000: Structure and Properties of Materials; ISBN: 979-8-203-16052-2

Website
Class content will be available on Brightspace. Your grades will also be posted there as well. (By the way, please disregard any calculations performed by Brightspace.)

Put These Examination Dates on Your Calendar!

Exam 1:	**Date & Time (TBD),**	**Location (TBD)**
Exam 2:	**Date & Time (TBD),**	**Location (TBD)**
Final Exam:	**Date & Time (TBD),**	**Location (TBD)**

CLASS PROCEDURES

Lectures will emphasize the most important points but you are also responsible for the details in the reading that time may not permit covering in lecture. Feel free to ask short questions in lecture. I am realistic enough to know that most of you will not read before class. However, the weekly assignments, that are due on Sunday night by 11:59 PM, encompass the important readings if you want to look at this beforehand.

Lectures will be given from an iPad which allows the instructor to annotate slides during lecture. Thus, the pdf of the lectures will make available to the students so that students can either print them off or annotate them using their preferred method (iPad, Surface Pro, etc.) The provided slides are only "sketches" of the lectures, and you will need to add your own notes to them during class – thus, hopefully, we will avoid "death by Power Point". *A bound copy of the notes will be available at CopyMat Services in West Lafayette for about $25. If you are in Indy taking the course please work with the MSE 230 class coordinator to obtain a copy.*

Recitations will consist of discussion led by the recitation instructor. Coverage will include past important homeworks, review of weekly assignments, and quizzes (that will occur approximately every other week). Questions about how your quiz is graded should be directed first to the TA. If the matter cannot be resolved between the two of you, then contact the instructor.

Quizzes (20 minute, closed-book) will cover basic concepts from the reading, lecture, recitation, and homework suggested for that week. Where needed, equations will be provided. There will be 5 quizzes this semester. Quizzes will be given at the end of recitation. Remind your TA 25 mins before classes end.

Expected format for quizzes

1. Neat, legible, organized, and in sufficient detail to demonstrate that you understood and performed the calculations yourself. Draw a system schematic as appropriate. Put problems in sequential order! Correct answers that just appear will not be given any credit.
2. Check the dimensions in any calculation and also that your answer makes physical sense.
3. Use reasonable precision in reporting the numerical value of calculations (e.g., not all 10 places of your calculator display) and use units consistently (SI unless otherwise given).

WEEKLY ASSIGNMENTS

In the content section on Brightspace you will find a "weekly assignments" folder. There is a weekly ***assignment due every week on Sunday by 11:59 EST pm*** (except for the last two weeks of the semester). You will have these weekly assignments even during exam weeks. The purpose of these weekly assignments is to get you in the book, and to serve as homework for the class. You should receive a perfect grade on these assignments if you are diligent. The scores are automatically transferred to Brightspace. I cannot give you extra time for these if you miss one even if you have a valid excuse; the points from missing day will be a small percentage of all the points you accrue throughout the semester. Also, you can always work ahead in this course. **Please note that you need to enter into these weekly assignments through Brightspace, following the "weekly assignments" folder.**

EXAMINATIONS

All examinations are closed-book but you may use up to 1 sheet of HANDWRITTEN notes (8.5" x 11", two sides). **You cannot use an iPad to create the crib sheet.** Besides this crib sheet, the only other things allowed for the exams are writing implements, eraser, straight edge (ruler), and a calculator. *You cannot use your phone as a calculator.* ***This year we are using multiple choice exams.*** More later on this. IF YOU GET EXTRA TIME FOR QUIZZES OR EXAMS YOU NEED TO LET US KNOW THE FIRST WEEK. Copy me and your TA on this information.

Make-up exams or quizzes *will be given only for the following verifiable reasons: serious illness, family emergencies, direct conflict with another scheduled exam (must inform instructor no later than two weeks prior), or official university absence. Do not schedule travel without knowing the final exam schedule.*

Grading

Exams (25% ea.)	50%
Final Exam (comprehensive)	30%
Weekly Assignments	10%
Quizzes (5 scores)	10%

Final grades

The C/B division will be no higher than the median score. The B/A and D/C divisions will depend on class performance, but will be no higher than the 80th and 20th percentile, respectively. Borderline cases may be raised up (but not down) based on subjective appraisal of the student's participation. If your average is below 40% you will have earned an F in this class.

OTHER COURSE INFORMATION AND CLASS HACKS

- The course overview page at the end of this syllabus is your friend. Use it to organize your life. Because all assignments are posted, you can always work ahead in this course.
- **Any** email sent to your TA should just include him or her. **Any** email sent to the instructor should also include your TA unless it regards an issue with your TA.
- Three exams total in MSE 230 (including the final exam). 80% of your grade comes from the exams. You can get a perfect weekly assignment grade (this is 10% of your overall grade).
- **98%** or more of what you will be tested over (includes exams and quizzes) will come from lectures/notes, recitation, and weekly assignments. Focus your efforts on this material to do well in this class.
- Exams in MSE 230 will be multiple choice. More information to follow.

Cheating

Academic Dishonesty
Purdue University Regulations, Part 5, Section III-B-2-a describes the formal policies governing academic dishonesty. A guide providing specific examples, tips, and consequences is available from the Office of the Dean of Students at http://www.purdue.edu/ODOS/osrr/integrity.htm.

The MSE 230 teaching staff will diligently monitor academic dishonesty in quizzes and exams. Students found to engage in academic dishonesty are subject to sanctions. For a first offense, the student will receive a zero score for the quiz or exam. Depending on the severity of the offense, the case may be referred to the Office of the Dean of Students for further action. A second offense will result in a failing grade for MSE 230 and definite referral to the Office of the Dean of Students.

Campus Emergency Policy
In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances. Any such changes will be posted to the course website. If you are unable to use Brightspace from home please let us know early in the semester so we can make other arrangements for your special needs.

THE BEST WAY TO DO WELL IN MY CLASS IS TO ATTEND ALL LECTURES. THE NOTES ARE KEY.

The next page is the class schedule. Some highlights:

- You have a weekly assignment **due every Sunday night by 11:59 EST pm**. These assignments are linked to your textbook and will be automatically graded and imported into Brightspace. For example, **Weekly Assignment 1 is due January 18 at 11:59 EST pm**. I suggest you not wait until Saturday night to do these. I cannot give extensions on this homework.
- Note that the total points of each assignment are indicated. For example, weekly assignment 1 is worth 96 of the total **1097 pts** (the sum of all the weekly assignments). If you miss this one, it costs you approximately 1 pt. from your final grade.
- The number of points for each weekly assignment is somewhat proportional to the time required to do the assignment.
- You have weekly assignments **due every week**, even during exam weeks. The only exceptions are: Spring Break week (3/13 – 3/17) and the weeks of 4/20 and 4/27.
- The weekly assignments will guide you through the **important readings of the textbook**.
- You should get a perfect grade on every week assignment if you are diligent.

Week	Meet	Date	Topics	Reading/Homework Weekly Assign.	Quiz or Exam Coverage
	L1	Jan 12	Introduction		
1	L2	Jan 14	Atom Attractions	Assign 1 (96 pts)	
	R1	J 15/16	Miller Indices Practice		
	H	Jan 19	MLK (no class)		
2	L3	Jan 21	How Atoms Self-Organize - Crystals	Assign 2 (114 pts)	
	R2	J 22/23	**Quiz 1**		**L2, L3**
	L4	Jan 26	Microstructure and X-ray characterization		
3	L5	Jan 28	Introduction to Deformation of Materials	Assign 3 (123 pts)	
	R3	J 29/30	Example Problems/Content Review/Q&A		
	L6	Feb 2	Why aren't Materials as Strong as expected	Assign 4 (49 pts)	
4	L7	Feb 4	How to Strengthen a Metal		
	R4	F 5/6	**Quiz 2**		**L4, L5**
	L8	Feb 9	Fracture Mechanics: role of cracks in failure	Assign 5 (85 pts)	
5	L9(A,B)	Feb 11	Materials Fatigue; Diffusion Introduction		
	R5	F 12/13	Example Problems/Content Review/Q&A		
	L10	Feb 16	How Atoms Move: Diffusion	Assign 6 (115 pts)	
6	L11	Feb 18	Simple Phase Diagrams		
	Ex 1	**Feb 16**	**Various Locations**	**6:30 – 7:30 pm**	**L1-L9A**
	L12	Feb 23	More Complex Phase Diagrams	Assign 7 (59 pts)	
7	L13	Feb 25	Fe-Fe_3C Phase Diag./Non-Equil. Micro.		
	R6	F 26/27	Review Exam 1		
	L14	Mar 2	There is More to Steel than Pearlite	Assign 8 (101pts)	
8	L15	Mar 4	More Non-Equilibrium Microstructures	Assign 9 (30 pts)	
	R7	M 5/6	**Quiz 3**		**L11 – L13**
	L16	Mar 9	Specialty Metals		
9	L17	Mar 11	Introduction to Ceramics	Assign 10 (71 pts)	
	R8	M 12/13	Example Problems/Content Review/Q&A		
	H	Mar 16	**Spring Break**		
10	H	Mar 18	**Spring Break**		
	H	M 19/20	No Recitation		
	L18	Mar 23	Overview of Ceramics Processing		
11	L19	Mar 25	Introduction to Polymers	Assign 11 (43 pts)	
	R9	M 26/27	**Quiz 4**		**L14– L16**
	L20	Mar 30	Properties of TP, TS and elastomer polymers		
12	L21	Apr 1	Composites	Assign 12 (114 pts)	
	R10	A 2/3	Example Problems/Content Review/Q&A		
	L22	Apr 6	Thermal Properties of Materials		
13	L23	Apr 8	Electrical Properties of Materials	Assign 13 (46 pts)	
	Ex 2	**Apr 9**	**Various Locations**	**8:00 – 9:00 pm**	**L9 (Diff.)-L18**
	L24	Apr 13	Band Structure of Materials		
14	L25	Apr 15	Semiconductors I	Assign 14/15 (51 pts)	
	R11	A 16/17	**Review Exam 2**		
	L26	Apr 20	Semiconductors II	No Assignment Due	
15	L27	Apr 22	Special Topic I: Nanomaterials synthesis		
	R12	A 23/24	**Quiz 5**		**L23-27**
	L28	Apr 27	Special Topics II: Nanomaterials properties	No Assignment Due	
16	L29	Apr 29	Special Topics III: Magnetic materials		
	R13	A30/M1	Example Problems/Content Review/Q&A		
17	**Final**	**TBD**			**L1-L29**